JAN–I–9

31 MAY 1944

JOINT ARMY-NAVY SPECIFICATION

INSULATORS, GLASS, RADIO

Army Number
71–4932

Navy Number
16–I–6

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and Navy, and shall, unless otherwise directed, be used in lieu of the following specifications:

U. S. Army

Bureau of Ships
Specification
16–I–6(Int)
1 Feb. 1944

A. APPLICABLE SPECIFICATIONS.

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–C–5–Capacitors, mica-dielectric, fixed.
JAN–I–10–Insulating material, ceramic, radio, Class L.

NAVY DEPARTMENT SPECIFICATIONS

General Specifications for Inspection of Material.[1]
39P16—General specification for packaging and packing for overseas shipment.
46B6—Brass, Naval, rolled: bars. plates, rods, shapes, sheets and strips.
46B8—Bronze, valve (composition M): castings.
46B10—Brass, Naval (composition N–c): castings.
46–I–8—Iron, malleable: castings.
46S17—Steel, cold-rolled or cold-drawn (screw stock).

B. TYPES.

B–1. Glass radio insulators covered by this specification shall be furnished in the types shown on figures 6 to 14, inclusive, as specified in the contract or order. (See par. H–2.)

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—The material for each part shall be as hereinafter specified. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

[1] Applicable only to Navy purchases.

All amendments to this specification are printed in the back of this specification and become a part thereof.

All amendments are to be printed face and back, on white 32 sub. sulphite, with continuing pagination at the back of the specification.

All specifications are to be perforated on the fold and 1 side stitch placed between the 2 groups of 3 holes, trim 3 sides.

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C–1a. Glass used in the construction of insulators shall be an approved grade conforming to the requirements of Joint Army-Navy Specification Jan–I–10. Where a definite grade of glass is not specified grade L–3 or better shall be used.

C–1b. Where a definite material for the hardware is not specified, a suitable material shall be used.

C–2. *Substitution of material.*—If the manufacturer desires to substitute for a specified or previously approved material or fabricated part, he shall submit to the bureau or agency concerned with his bid or sample a statement to that effect, describing the proposed substitution, together with evidence to substantiate his claim that such substitute is suitable. Similar notification and substantiating evidence shall be submitted at any later time if substitution becomes necessary or desirable. At the discretion of the bureau or agency concerned test samples may be required to prove the suitability of the proposed substitute.

C–3. *Screw threads.*—Where practicable, screw threads for all bolts, machine screws, nuts, and tapped holes shall be of National Form in accordance with the Screw-Thread Standards for Federal Services, Handbook H28, except as modified in Table IX for glass. Unless otherwise specified, screws smaller than size 6–32 will not be permitted.

C–4. *Workmanship.*—Insulators shall be manufactured and processed in a careful and workmanlike manner in accordance with good design and practice.

D. GENERAL REQUIREMENTS.

D–1. *Definitions.*—

D–1a. *Holes.*—

D–1a (1). *Mounting holes.*—Mounting holes are defined as those holes in the insulator which are used as points of support for the insulator or for other members of an assembly.

D–1a (2). *Other holes.*—Other holes are defined as all holes other than mounting holes, such as winding or tap-off holes in coil forms, lead holes, perforations for ventilation or for weight reduction purposes, etc.

D–1b. *Surface marks.*—The following surface marks are sometimes found in glass insulators:

D–1b (1). *Blister.*—

D–1b (1) a. A bubble or void in glass of characteristic A, B, or C is known as a blister if it is larger than $\frac{1}{16}$ inch in at least 1 dimension. It can be opened or buried. (See par. D–1b (8).)

D–1b (1) b. Blisters in ware of characteristic D are small bulges or lumps. They do not necessarily indicate the presence of a cavity under the surface.

D–1b (2). *Check.*—An incipient crack is known as a check. It is usually considered as being limited to a single surface or a corner, not extending across or through the wall section. Checks may be caused in the fabricating operation, in grinding or cutting, from a bruise, or from thermal shock.

D–1b (3). *Chip.*—A place on the surface where a piece of glass has been completely broken out is known as a chip.

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D–1b (4). *Cord.*—Cord is defined as a visible streak in glass resulting from the inclusion of a band of glass which is not homogeneous with the remainder, and consequently has different optical and other properties.

D–1b (5). *Fin.*—A thin projection of glass resulting from imperfect matching of the mold joints is known as a fin.

D–1b (6). *Lap.*—A lap is defined as an irregularity on a glass surface caused by the glass folding over while the glass is still in the viscous state and being pressed in the mold.

D–1b (7). *Mold marks.*—Marks on glass surfaces transferred from scratches or other imperfections on the mold surfaces are known as mold marks.

D–1b (8). *Seed.*—A seed is defined as a small bubble in glass and may appear as a collection of bubbles. A seed is usually considered to be not larger than $\frac{1}{16}$ inch in any dimension. (See par. D–1b (1) a.)

D–1b (9). *Shear mark.*—A shear mark is defined as a surface imperfection resulting from the severing of the body or "gob" of glass loaded into the press mold. All pressed pieces of glass will contain shear marks.

D–1b (10). *Stone.*—A stone is defined as a nonvitreous inclusion in glass. Stones may result from undissolved batch material, or from contamination from refractory furnace lining.

D–1c. *Annealing.*—The process of gradually cooling glass pieces between certain temperature limits and at certain minimum rates of cooling so that residual strains in the glass will be relieved, is known as annealing.

D–1d. *Camber.*—For the purpose of this specification, camber is defined in percent as the ratio $100\frac{e}{d}$ where e is the maximum perpendicular distance from the chord to the corresponding curved surface and d is a reference length measured between the sector extremities. (See figs. 2, 3, and 4 and par. E–4e (2) c.)

D–1e. *Siding.*—Siding is variation in wall thickness in any one plane at right angles to the longitudinal axis of the piece.

D–2. *Condition of surfaces.*—Surfaces which are chipped or blistered to the extent that the life and usefulness of the insulator are impaired shall be cause for rejection of the insulator. Rough or irregular edges caused by minute chips or nicks are not considered as impairing the life and usefulness of the insulator unless sharp edges are specified on the applicable figure. The following information shall be used in determining the significance of all surface imperfections:

D–2a. *Chips, blisters.*—Insulators having surface flaws shall be rejected in the event that such imperfections result in:

(1) Surfaces surrounding mounting holes which will not withstand the stress of mounting or will not provide secure mounting.

(2) Bearing surfaces which will not withstand bearing pressure or friction of associated assembly details.

(3) Surfaces of grooves or other indentations provided for carrying coil windings or other wiring which will not provide adequate support for winding or which will damage the wire.

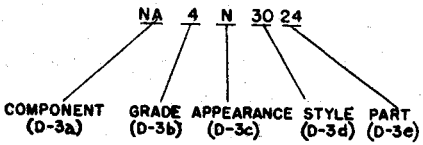
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(4) Sealing surfaces which will not provide sufficient support for materials used in sealing inserts or associated assembly details.
(5) Glazed surfaces which are not substantially free from chips occurring after glazing.
(6) Blisters which might easily become open cavities.
(7) Blisters which are excessively large or occur frequently on any surface so as to impair life and usefulness of the insulator.

NOTE.—It is deemed impractical to set up even general standards of quality for size and quantity of chips or blisters inasmuch as either must be judged in conjunction with the other. Since both size and application of ceramic parts are subject to wide variation from one piece to another, it is believed that questions of size and quantity of chips or blisters should be considered on a common-sense basis rather than by any empirical formula covering "number of chips or blisters per square or linear inch" or "loss in volume."

D–2b. *Surface marks.*—All insulators having surface marks shall be rejected in the event that the following conditions exist:
(1) Where there is evidence of a cavity.
(2) Where they occur frequently or are excessively large so as to impair life and usefulness of the insulator.
(3) Where they occur between coil-winding grooves and cover more than one-half the space between the grooves.

D–3. *Type designation.*—The type designation for glass insulators is as follows:



D–3a. *Component.*—Glass insulators are identified by the symbol "NA", where "N" indicates "insulator" and "A" indicates "glass."

D–3b. *Grade.*—The grade of insulating material used is identified by a 1-digit number corresponding to the grades covered by Joint Army-Navy Specification JAN–I–10.

D–3c. *Appearance.*—The appearance specified is identified by a single letter which differentiates between clear, opaque, and translucent insulators as follows:

Appearance:	Designating letter
Clear	C
Opaque	N
Translucent	W

D–3d. *Style.*—The style designation is a 2-digit number appearing on individual figures and identifying the general form of the insulator. Where a style number is less than 10, it shall be preceded by a zero to avoid confusion with the part number. (See par. D–3e.)

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D–3e. *Part.*—The part number is a 2-digit number identifying individual insulators shown on individual figures. Where practicable, as in the case of pillar and strain insulators, the part number will be based on the length of the part expressed in eighths of an inch.

E. DETAIL REQUIREMENTS.

E–1. Insulators shall be manufactured in accordance with the requirements of the applicable figure.

E–2. *Fittings.*—Insulators shall be furnished with such fittings as may be specified on the applicable figure or in the contract or order. (See par. H–2.) The methods employed for securing the fittings shall be of such nature as to insure that the completed insulator will withstand the specified tests.

E–3. *Tapped holes.*—Unless otherwise specified, all tapped holes in the glass shall be slightly countersunk.

E–4. *Tolerances.*—Unless otherwise specified, the following tolerances will be permitted on all dimensions of insulators:

E–4a. *Holes in characteristic A, B, or C ware.* (*See par. 2 of appendix.*)

E–4a (1). The tolerance on holes in these classes shall be as shown in Table I.

TABLE I

Diameter (inch)	Tolerance (±)	
	Drilled or pressed	Punched
	Inch	*Inch*
From 1/8 to less than 1/4	1/64	0.010
From 1/4 to less than 1	1/32	----------

E–4a (2). For insulators fabricated with the use of jigs, the tolerance on hole-center spacing shall be as shown in Table II.

TABLE II

Method	Tolerance on hole-center spacing (±)
	Inch
Drilled or pressed	1/32
Punched	0.020

E–4b. *Characteristic A ware.* (*See par. 3 of appendix.*)—

E–4b (1). Draft on inside shall be not greater than the value shown in Table III.

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TABLE III

Depth (inches)	Draft
	Degrees
Less than 3	2
From 3 to less than 5	3
From 5 to 7, inclusive	4

E–4b (2). *Diameter.*—The tolerance on diameter of this ware, including "out-of-round," shall be not greater than the values shown in Table IV.

TABLE IV

Diameter (inches)	Tolerance (±)[1]
	Inch
Less than 4	1/32
From 4 to less than 6	3/64
6 or greater	1/16

[1] These tolerances apply also to solid, non-cavity ware.

E–4b (3). *Side-wall thickness.*—Side walls that are approximately vertical shall have a side-wall thickness tolerance of ± 1/32 inch, exclusive of taper.

E–4b (4). *Bottom thickness.*—The tolerance on closed-end thickness shall be not greater than ± 50 percent, unless otherwise specified. This variation will result in a corresponding change in the inside depth of the piece.

E–4b (5). *Over-all height or length.*—The tolerance on over-all height or length shall be ± 1/16 inch on mold-finished ware.

E–4c. *Characteristic B ware.* (*See par. 4 of appendix.*)—

E–4c (1). *Diameter—circular cross section.*—The tolerance on diameter of this ware, including "out-of-round," shall be as shown in Table V.

TABLE V

Diameter (inches)	Tolerance (±)
	Inch
Less than 3	0.025
From 3 to less than 6	.032
From 6 to less than 10	.047
10 or greater	.063

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E–4c (2). *Ellipticity.*—Ellipticity, or "out-of-round," shall be determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient shall not exceed 1.02.

E–4c (3). *Wall variation.*—For open-ended cylinders with a ratio of $\frac{\text{length}}{\text{diameter}}$ not greater than 5, the wall-thickness tolerance shall be ±33 percent; for a ratio greater than 5, the wall thickness tolerance shall be ±50 percent.

E–4c (3) a. With one end closed, the maximum thickness of the end shall be not more than 50 percent greater than the maximum dimension for the side-wall thickness.

E–4d. *Characteristic C ware.* (*See par. 5 of appendix.*)—

E–4d (1). *Outside diameter.*—

E–4d (1) a. *Machine-drawn and hand-drawn tubing.*—The tolerance on outside diameter of machine-drawn and hand-drawn tubing shall be not greater than that shown on figure 1. This tolerance includes allowance for ellipticity, or "out-of-round."

E–4d (1) b. *Solid rod.* (*See par. 5b (1) a of appendix.*)—The tolerance on outside diameter of hand-drawn solid rod shall be not greater than that shown for hand-drawn tubing in figure 1. For machine-drawn solid rod, the tolerance on outside diameter shall be not greater than that given by the formula:

$$\text{Tolerance} = \pm (0.035D + 0.015)$$

where D is the outside diameter of the solid rod, and all dimensions are in inches.

E–4d (2). *Wall thickness.*—The tolerance on wall thickness for machine-drawn tubing shall be ±25 percent of nominal wall thickness; and for hand-drawn tubing, ±33 percent of nominal wall thickness. These values include allowances for nonconcentricity of bore and outside diameter.

E–4d (3). *Camber.*—The maximum permissible camber for machine-drawn tubing and rod shall be 0.5 percent of the length of the tubing; and for hand-drawn tubing and rod, 0.75 percent of the length of the tubing.

E–4e. *Characteristic D ware.* (*See par. 6 of appendix.*)—

E–4e (1). *Noncritical dimensions.*—All noncritical dimensions, which are indicated as such by common fractional dimensions on applicable figures or in the contract or orders, shall have a tolerance of ±2 percent of the dimension involved, but not more than ±⅛ inch, on dimensions up to and including 12½ inches. On all dimensions greater than 12½ inches, a tolerance of ±1 percent will be permitted. In no case shall any tolerance be less than ±1/64 inch.

E–4e (2). *Critical dimensions.*—All critical dimensions, which are indicated as such by decimal dimensions on applicable figures or in the contract or orders, may have tolerances of ±1 percent of the dimension involved, but not less than ±0.010 inch, except as follows:

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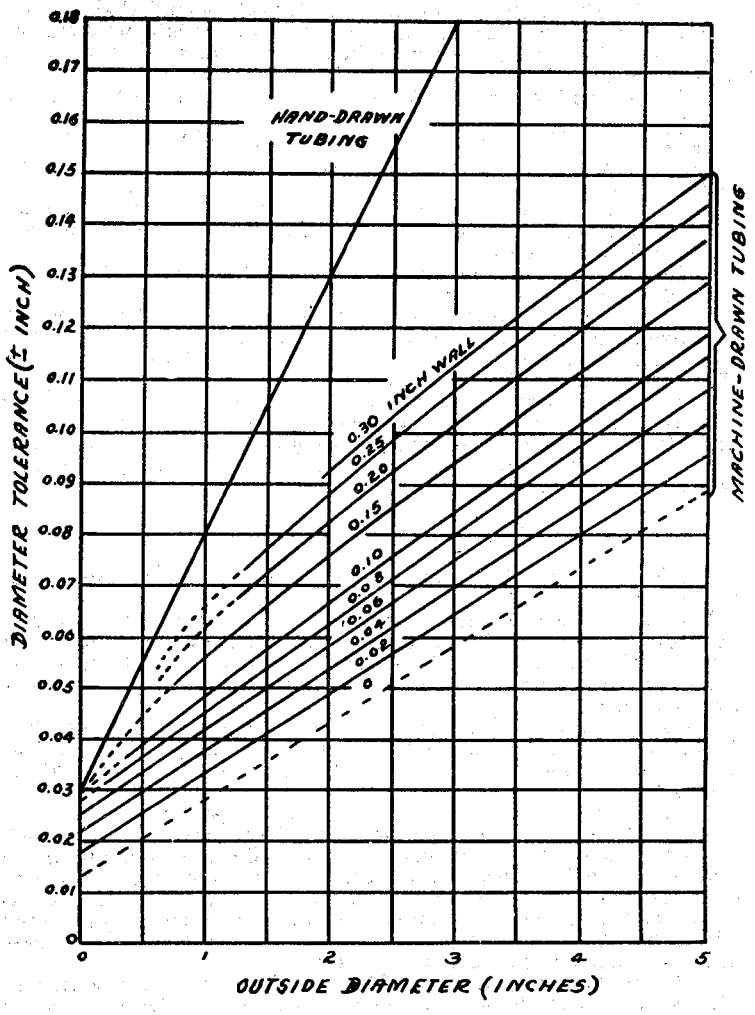


FIGURE 1.—TOLERANCES ON MACHINE-DRAWN AND HAND-DRAWN TUBING OF CHARACTERISTIC C.

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E–4e (2) a. *Cylindrical shapes—outside and inside diameters.*—A tolerance of ±2 percent of the dimension involved, but not less than ±0.010 inch, will be permitted on all diameters up to and including 1 inch. Where the diameter is greater than 1 inch, ± (1 percent of the diameter involved + 0.010 inch), shall be used. Unless otherwise specified, a minimum tolerance of ±0.010 inch will be permitted on nominal wall thickness dimensions.

Example:

Tube (*inches*)	*Tolerance* (±) (*inch*)
O. D. 0.500	O. D. 0.010
I. D. .125	I. D. .010
O. D. 1.125	O. D. .021
I. D. .825	I. D. .0165
O. D. 2.00	O. D. .030
I. D. 1.750	I. D. .0275

E–4e (2) b. *Lineal dimensions.*—All lineal dimensions representing over-all lengths shall have tolerances as shown in Table VI.

TABLE VI

Length (inches)	Tolerance on length (±)
	Inch
Less than 0.500	0.010
From 0.500 to less than 0.750	.012
From 0.750 to less than 1.000	.015
From 1.000 to less than 1.250	.020
From 1.250 to less than 1.500	.022
From 1.500 to less than 2.000	.025
From 2.000 to less than 2.500	.030
From 2.500 to less than 3.000	.035
From 3.000 to less than 4.000	.040
From 4.000 to less than 5.000	.045
From 5.000 to less than 6.000	.050
6.000 or greater	.055

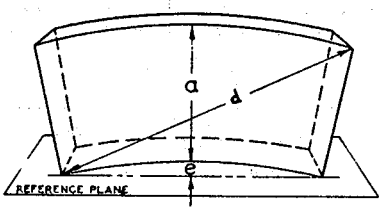
E–4e (2) c. *Camber.*—The camber (see par. D–1d) shown in Table VII shall apply to all surfaces of all shapes. Figures 2, 3, and 4 shall be used to establish the ratios listed in Table VII. For flat shapes, disks, plates, and the like, these limits shall apply only where the thickness is equal to or greater than the thickness determined by the formulas given in paragraph 6a (10) of the appendix. Where the thickness is less than that determined from paragraph 6a (10) of the appendix, the camber shall be specified on the applicable figure or in the contract or order (see par. H–2):

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TABLE VII

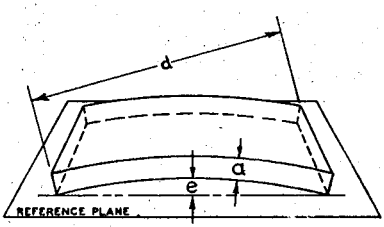
$\frac{d}{a}$	Maximum camber (percent of d) to be applied to surfaces
Less than 2	0.25
From 2 to less than 3	.30
From 3 to less than 5	.35
From 5 to less than 8	.40
From 8 to less than 13	.45
From 13 to less than 20	.50
From 20 to less than 30	.55
From 30 to less than 40	.60
From 40 to less than 50	.65

NOTE.—In determining "maximum camber" permitted, values of e (figs. 2, 3, and 4) less than 0.006 inch shall be disregarded, unless otherwise specified.



d = Maximum diagonal length on the face
a = Minimum dimension in same direction as e

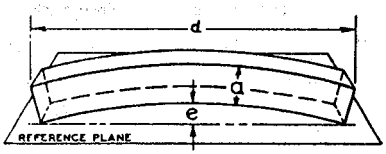
FIGURE 2.



d = Maximum diagonal chord length
a = Minimum dimension in same direction as e

FIGURE 3.

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Square or circular section

d = Maximum length of piece
a = Minimum diameter or minimum dimension in same direction as *e*

FIGURE 4.

E–4e (2) d. *Ellipticity of tubes.*—The ellipticity shall be determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient shall not exceed 1.02 and shall apply to cylindrical shapes having a wall thickness of at least 10 percent of the outside diameter. If the wall thickness at the thinnest section of the wall is less than 10 percent of the outside diameter, the same ratio shall not exceed 1.03.

E–4e (2) e. *Hole centers.*—Tolerances of ±0.005 inch for the dimensions between centers of holes for all hole-center spacings less than 0.500 inch, and ± 1 percent of the dimension involved for all hole-center spacings 0.500 inch and greater, will be permitted.

NOTE.—If pin gages are used for inspecting hole-center spacing, the design of the gage should be such as to meet the specific requirements. In using pin gages consideration should be given to the tolerance between hole centers as well as hole-diameter tolerance.

E–4e (2) f. *Hole diameters.*—Tolerances of ±0.005 inch for all hole diameters up to 0.500 inch, and ±1 percent of the diameter for all hole diameters 0.500 inch and greater, will be permitted. Holes which are not perfect circles shall have the same tolerance applied to the minor axis only.

E–4e (2) g. *Parallel limits.*—Parellelism of surfaces will be considered satisfactory if the thickness measured at any point is within the established thickness tolerance of the piece.

E–4f. *Tapped holes.*—Tapped holes shall be within the limits shown in Table IX when tested as described in paragraph F–5b.

E–5. *Thermal shock.*—When specified on the applicable figure or in the contract or order (see par. H–2), insulators shall be capable of withstanding the thermal shock test described in paragraph F–6. The number of failures shall not exceed those permitted in Table VIII.

E–6. *Mechanical load.*—When specified on the applicable figure or in the contract or order (see par. H–2), insulators shall be capable of withstanding the mechanical load specified in paragraph F–7.

E–7. *High potential.*—When specified on the applicable figure or in the contract or order (see par. H–2), insulators shall be capable of withstanding the high-potential test voltages specified in paragraph F–8.

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E–8. *Galvanizing (zinc coating).*—Galvanizing on metal parts of insulators shall meet the Preece test requirements.

E–9. *Other requirements.*—If additional requirements are given on the applicable figure, or in the contract or order (see par. H–2), insulators shall be capable of complying with the specified tests. (See par. F–10.)

E–10. *Marking.*—

E–10a. *General.*—Each insulator shall be permanently marked with proper type designation (see par. D–3) and the manufacturer's name or symbol. Such marking shall be accomplished by means of stamping or molding all necessary markings on the body of the insulator, except as specified in paragraph E–10b.

E–10b. *Marking of cross sections.*—Items such as coil tubes, which are standardized only with respect to cross section and which are not required to carry individual part numbers, may be marked with the component and grade symbols of the type designations. (See par. D–3.)

Example: NA3.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Test equipment and inspection facilities.*—The insulator manufacturer shall furnish and maintain all necessary facilities and equipment for making all tests and inspection which are made at the place of manufacture and shall perform all tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity to enable the inspection to keep up with production. The test equipment shall be of sufficient accuracy and quality to permit performance of the required tests. Where possible, the inspection of insulators shall be at the place of manufacture.

F–2. *Special gages.*—If special gages for inspection of parts are to be used by the bureau or agency concerned, requirements for such gages or a set of gages themselves, if practicable, should be furnished to the manufacturer by the bureau or agency concerned.

F–3. *Production tests.*—Each insulator shall be subjected to visual inspection and to other tests as specified on the applicable figure or in the contract or order. (See par H–2c.) Failures on production tests shall result in the rejection of the item failing, unless it can be shown to the satisfaction of the Government inspector that the items can be corrected with respect to the substandard features, in which case the items shall be corrected and reinspected.

F–4. *Production-sampling tests.*—For production sampling, insulators shall be selected by the Government inspector from insulators that have passed production tests. Table VIII lists the quantities of insulators to be selected. All tests, except visual inspection, shall be made only when specified. Insulators shall be subjected to the applicable tests, in the order shown, and there shall be no more failures than the number specified. If the number of failures exceeds the maximum number permitted in Table VIII a retest shall be made. If the number of failures of the same type still exceeds the maximum allowable, this shall not necessarily be cause for complete rejection. If the failures on

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the retest exceed 2 percent of the entire lot, the entire lot shall be subjected to individual tests, and the items failing shall be rejected. If the number of failures does not exceed 2 percent, the lot shall be accepted, unless, in the opinion of the Government inspector, the nature of the defect is detrimental to the life or performance of the part, in which case the entire lot shall be subjected to individual inspection.

TABLE VIII

Test	Test requirements	Test procedure	Quantity in continuous production				
			0–50	51–100	101–500	501–5,000	Each additional 5,000
			A B	A B	A B	A B	A B
Visual inspection	D–3, E–1, E–2, E–3, E–4, E–10	F–5	0 0	4 0	6 0	12 0	12 0
Thermal shock	E–5	F–6	0 0	4 1	6 1	12 2	12 2
Mechanical load	E–6	F–7	(1)	(1)	(1)	(1)	(1)
High potential	E–7	F–8	(1)	(1)	(1)	(1)	(1)
Galvanizing	E–8	F–9	0 0	4 0	6 0	12 0	12 0
Other	E–9	F–10	(1)	(1)	(1)	(1)	(1)

[1] As specified.

Where:

A = number of samples required for test.

B = maximum number of failures permitted.

F–5. *Visual inspection.*—Insulators shall be inspected to ascertain that the workmanship is satisfactory and that the quantity is correct.

F–5a. *Tolerances.*—Insulators shall be inspected to ascertain that their physical dimensions are within the specified limits. (Where specific tolerances are not given on the applicable figures, the general tolerances as specified in paragraph E–4 shall apply.)

F–5b. *Inspection of tapped holes.*—The following gages shall be used for the inspection of threads and tapped holes:

(1) For checking the minor diameter, *d* (see fig. 5): A plain Go gage pin, diameter A, and a plain Not Go gage pin, diameter B.

(2) A Go thread plug gage.

NOTE.—See Table IX for gage dimensions.

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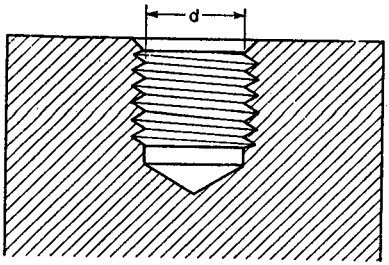


FIGURE 5.—INSPECTION OF TAPPED HOLES.

TABLE IX

Size of tapped hole	Column 1	Column 2	Column 3	Threads per inch	Acceptable gaging combinations
	Go gage pin diameter A	Not Go gage pin diameter B	Go thread plug gage		
	Inch	*Inch*	*Inch*		
4–40 [1]	0.082	0.098	0.112	40	Columns 1, 2, and 3 or Columns 2 and 3 only.
6–32	.104	.122	.138	32	
8–32	.130	.148	.164	32	
10–32	.157	.174	.190	32	
¼–20	.195	.222	.250	20	
5/16–18	.252	.281	.312	18	
⅜–16	.307	.337	.375	16	

[1] Where specific approval has been obtained for the use of 4–40 tapped holes, these limits shall apply.

F–6. *Thermal shock test.*—Sample insulators shall be immersed in an ice-water bath for 10 minutes and then transferred as quickly as possible to a hot-water bath and allowed to remain there for 10 minutes. The samples shall be transferred back to the ice-water bath and the thermal cycles shall be continued until each sample breaks or has successfully withstood the required number of cycles. Unless otherwise specified, the test specimens shall be exposed to 20 thermal cycles. The hot-water bath shall be maintained at not lower than 94° C. (201° F.) and the cold-water bath at not higher than 4° C. (39° F.).

F–7. *Mechanical load test.*—Insulators shall be tested to determine their ability to withstand the type and magnitude of loading as specified on the applicable figure or in the contract or order. (See par. H–2c.)

F–8. *High-potential test.*—Insulators shall be subjected to a high-potential test to determine their ability to withstand without breakdown the voltage specified on the applicable figure or in the contract or order (see par. H–2c).

F–9. *Galvanizing (zinc coating).*—Metal parts of insulators shall be tested for adequate galvanizing by subjecting them to the Preece test.

[JAN–I–9]

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F–10. *Other tests.*—Insulators shall be subjected to such additional tests as may be required on the applicable figure or in the contract or order. (See par. H–2c.)

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Army.*—Shipments of the subject equipment shall be packaged, packed, and marked as specified in the bid request, contract, or basic equipment specification.

G–2. *Navy.*—

G–2a. *Packaging.*—

G–2a (1). *Individual shipment.*—Unless otherwise specified, the subject commodity, when shipped as replacement parts, individual units, or as kit groups, shall be individually packaged in a reverse tuck, straight tuck, or two-piece telescope style folding carton. The minimum caliper of the board shall be 0.016 inch. Each box shall contain one insulator and shall be designed to fit the insulator in a compact manner.

G–2a (2). *Bulk shipment.*—Unless otherwise specified, the subject commodity when shipped in bulk shall require no packaging.

G–2b. *Packing.*

G–2b (1). *For domestic shipment.*—

G–2b (1) a. *Individual shipment.*—Unless otherwise specified, the subject commodity, packaged as described in paragraph G–2a (1), shall be packed in corrugated boxes conforming to the following requirement :

Gross weight of cartons (max.)	Average bursting strength (min.)	Sum of inside dimensions (min.)
Pounds	*Pounds*	*Inches*
40	200	60
65	275	75
90	350	90

For inspection purposes, all flaps shall be securely sealed in such a manner to permit opening and subsequent reclosure without damage to the upper flaps.

G–2b (1) b. *Bulk shipment.*—Unless otherwise specified, insulators shall be packed in corrugated boxes employing cell partitions. Each partition cell shall be designed to fit one insulator compactly. When two or more tiers of partitions are used per container, dividing or separating pads shall be placed between each tier. The boxes and interior packing shall conform to the requirements specified in paragraph G–2b (1) a.

G–2b (2). *For overseas shipment.*—Unless otherwise specified, the subject commodity packaged as described in paragraph G–2a (1) and packed as described in paragraph G–2b (1), shall be overpacked in cleated plywood boxes or nailed wood boxes conforming to the requirements of Sections V and VI, respectively, of Navy Department Specification 39P16, except that in cleated plywood boxes no cleat drainage ducts will be required. Each box shall be lined with a sealed waterproof bag or its equivalent made in conformance with the con-

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struction and tests of Section XXV. The material used shall have a minimum water resistance of 15 hours. Each box shall be strapped in accordance with the requirements of the applicable section of Navy Department Specification 39P16. The gross weight shall not exceed approximately 150 pounds.

G–2c. *Marking.*—

G–2c (1). *Packages.*—Unless otherwise specified, packages shall be conspicuously marked on each of two adjacent sides in black with the following information completed:

Article_____
Type Number_____
Stock Number_____
Order Number_____
Inspection_____
Contractor_____
Manufacturer_____
Quantity_____

G–2c (2). *Shipping containers.*—Unless otherwise specified, shipping containers shall be conspicuously marked in waterproof black on two adjacent sides with the following information completed, as indicated:

Article_____
Type Number_____
Stock Number_____
Order Number_____
Quantity_____
Contractor_____
Manufacturer_____
Gross Weight_____
FRAGILE—HANDLE WITH CARE.

In addition to the foregoing, marking shall conform to the requirements of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should contain the title of the specification, the number, and date.

H–2. Requests, requisitions, schedules, and contracts or orders should cover the following features:

(*a*) Type of insulator desired.
(*b*) Desired fittings not shown on applicable figures.
(*c*) Any additional requirements or tests desired.
(*d*) Any special camber for shapes.

H–3. This specification is not intended to be restrictive with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered provided the bid clearly describes the specific points in which the construction proposed differs from that described herein, and provided further, that all performance tests and other requirements of this specification are met. When exceptions are not clearly described and indicated as such, it will be assumed that bidders are offering insulators in strict accordance with the requirements of this specification.

[JAN–I–9]

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H–4. This specification has been prepared through the coordinated efforts of representatives of industry and the War and Navy Departments.

H–5. Many additional sizes and shapes of glass radio insulators, not covered by this specification, are now in use by the Armed Forces. Existing designs under which equipment is being, and will continue to be fabricated, are such as not to permit complete substitution with insulators conforming to the requirements of this specification, neither are the requirements suitable in all cases for direct replacement of existing insulators.

H–6. This specification has been editorially rearranged but is identical in all essential technical respects to the November 29, 1943, issue of the American War Standards C75.8—1943. References to American War Standard C75.1—1943 and C75.3—1942 have been deleted and Joint Army-Navy Specifications JAN–I–10 and JAN–C–5, which are identical in all essential technical respects, are referenced. Paragraphs on approval of material have been deleted as approval is covered in the material specification.

H–7. Matters relating to the use of components furnished in accordance with the requirements of this specification for use in connection with the design, production, and procurement of all Naval Radio, Radar, and Underwater Sound Equipment on the Electronics Precedence List should be referred to the Bureau of Ships, Radio Division, Navy Department, Washington, 25 D. C.

H–8. Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–9. Copies of Joint Army-Navy specifications (required for Army purchases) may be obtained, as indicated in the "Index of United States Army and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of joint Army-Navy specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–10. Copies of Joint Army-Navy specifications (required for Navy purchases) and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Navy: ShAOS.
[JAN–I–9]

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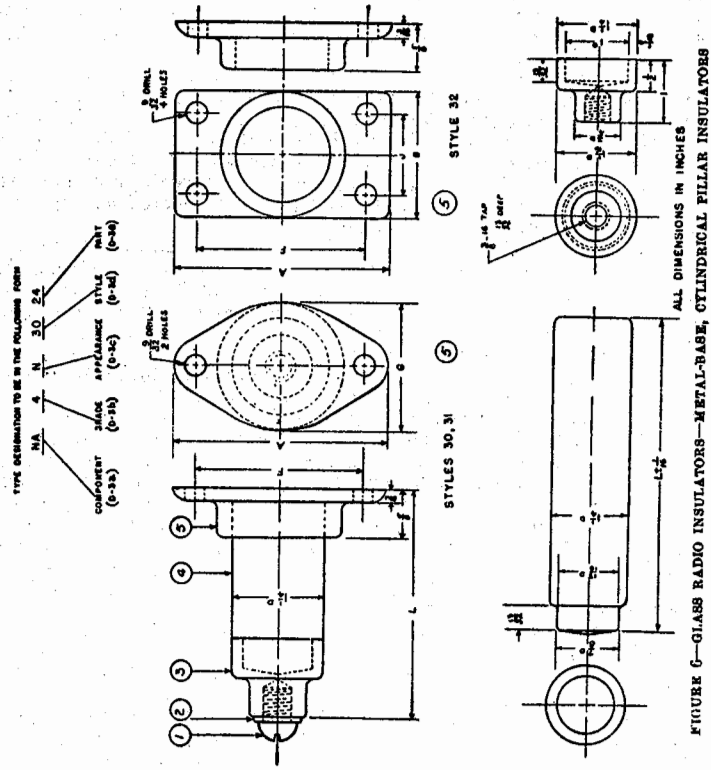


FIGURE 6—GLASS RADIO INSULATORS—METAL-BASE, CYLINDRICAL PILLAR INSULATORS

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Figure 6—Glass Radio Insulators—Metal-Base, Cylindrical Pillar Insulators—(Continued)

BILL OF MATERIALS

Part No.	Name of part	Quantity required	Remarks
1	Round-head machine screw	1	3/8-16 x ½, Brass, Navy Dept Specification 46B6.
2	Flat washer	1	1/16 thick, Brass, for part No. 1, Navy Dept Specification 46B6.
3	Cap	1	Brass, Navy Dept Specifications 46B6, 46B8, or 46B10.
4	Insulator	1	Glass, Joint Army-Navy Specification JAN-I-10.
5	Base	1	Malleable Iron, Navy Dept Specification 46-I-8; or Brass, Navy Dept Specifications 46B6, 46B8, or 46B10.

Tolerances on all dimensions of ceramic parts shall be as specified herein, unless otherwise indicated.
Bottom of base to be finished before galvanizing.
Samples shall be tested to determine compliance with E-8.
Top of cap and bottom of base shall be parallel to each other and perpendicular to center axis.
Method of securing metal parts to ceramic material optional.
For bolt-head clearance at mounting holes, the minimum radius of flat area measured from center of hole shall be ½ inch.

Type designation [1]	Dimensions (inches)					Insulators of following grades are available in appearances shown			
	L±1/16	A	F	G	J	L-3	L-4	L-5	L-6
NA--3024	3	3 1/16	2 1/4	1 1 1/16		C	N	CN	C
NA--3056	7	3 1/16	2 1/4	1 1 1/16		C	N	CN	C
NA--3080	10	3 1/16	2 1/4	1 1 1/16		C	N	CN	C
NA--3124	3	3 1/16	2 5/16	1 1 1/16		C	N	CN	C
NA--3156	7	3 1/16	2 5/16	1 1 1/16		C	N	CN	C
NA--3180	10	3 1/16	2 5/16	1 1 1/16		C	N	CN	C
NA--3224	3	2 7/8	2 1/4	1 1 1/16	1	C	N	CN	C
NA--3256	7	2 7/8	2 1/4	1 1 1/16	1	C	N	CN	C
NA--3280	10	2 7/8	2 1/4	1 1 1/16	1	C	N	CN	C

[1] Complete type designation will include additional letter symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (--).

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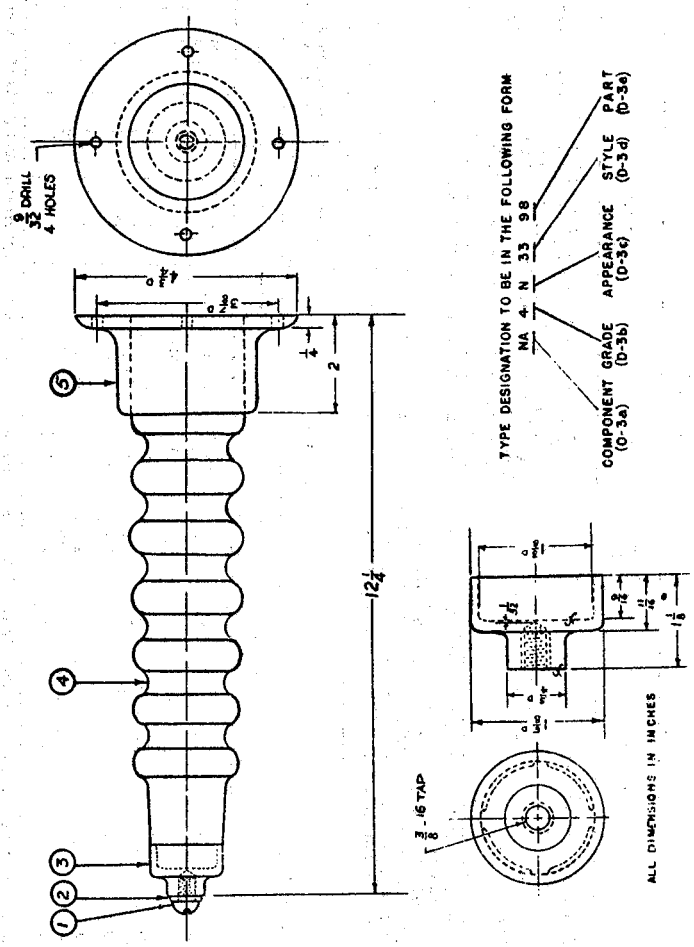


FIGURE 7.—GLASS RADIO INSULATORS—METAL-BASE, PILLAR INSULATORS

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Figure 7—Glass Radio Insulators—Metal-Base, Cylindrical Pillar Insulators—(Continued)

BILL OF MATERIALS

Part No.	Name of part	Quantity required	Remarks
1	Round-head machine screw	1	3/16-16 x 1/2, brass, Navy Dept Specification 46B6.
2	Flat washer	1	1/16 thick, brass, for part No. 1. Navy Dept Specification 46B6.
3	Cap	1	Brass, Navy Dept Specifications 46B6, 46B8, or 46B10.
4	Insulator	1	Glass, Joint Army-Navy Specification JAN-I-10.
5	Base	1	Malleable Iron, Navy Dept Specification 46-I-6; or Brass, Navy Dept Specifications 46B6, 46B8, or 46B10.

Tolerances on all dimensions of ceramic parts shall be as specified herein, unless otherwise indicated.
Bottom of base to be finished before galvanizing.
Samples shall be tested to determine compliance with E-8.
Top of cap and bottom of base shall be parallel to each other and perpendicular to center axis.
Method of securing metal parts to ceramic material optional.
For bolt-head clearance at mounting holes, the minimum radius of flat area measured from center of hole shall be 1/2 inch.

Type designation [1]	Insulators of following grades are available in appearances shown			
	L-3	L-4	L-5	L-6
Na-- 3398	C	N	CN	C

[1] Complete type designation will include additional letter symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (- -).

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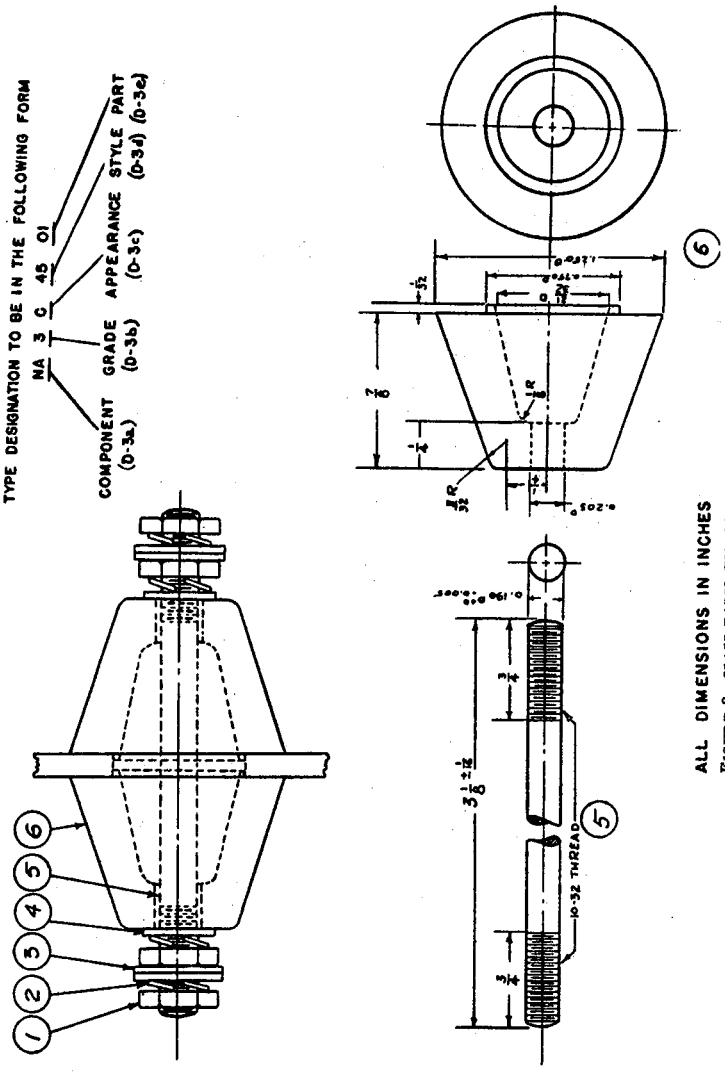


FIGURE 8—GLASS RADIO INSULATORS—LEAD-IN INSULATORS

[JAN-I-9]

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Figure 8—Glass Radio Insulators—Lead-In Insulators—(Continued)

BILL OF MATERIALS

Part No.	Name of part	Quantity required	Remarks
1	Hexagon nut	4	10–32, Brass, for part No. 5, Navy Dept. Specification 46B6.
2	Lock washer	4	Steel, for part No. 5, Navy Dept. Specification 46S17.
3	Flat washer	4	Brass, for part No. 5, Navy Dept. Specification 46B6.
4	Flat washer	2	⅛ thick, Lead, for part No. 5.
5	Rod	1	Brass, Navy Dept. Specification 46B6.
6	Insulator	2	Glass, Joint Army-Navy specification JAN–I–10.

Tolerances on all dimensions of ceramic parts shall be as specified herein, unless otherwise indicated.

Type designation [1]	Insulators of following grades are available in appearances shown			
	L–3	L–4	L–5	L–6
NA - - 4501	C	N	CN	C

[1] Complete type designation will include additional symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (- -).

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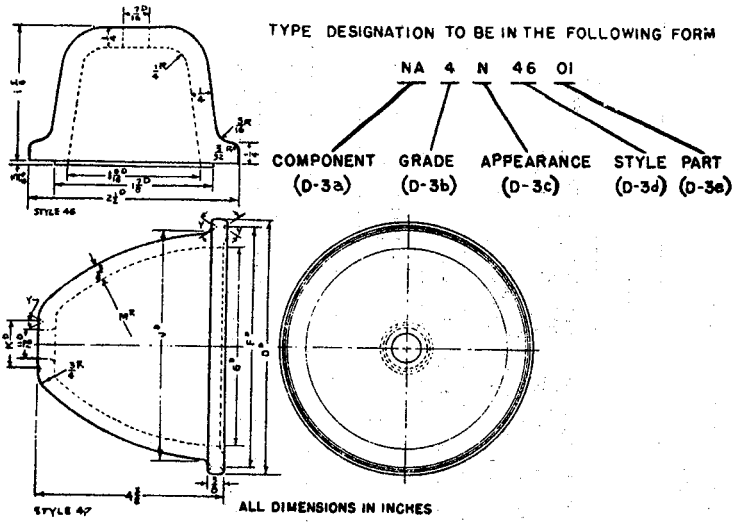


FIGURE 9—GLASS RADIO INSULATORS—LEAD-IN INSULATORS

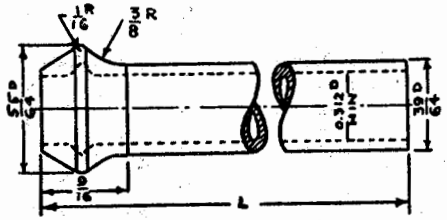
Tolerances on all dimensions shall be as specified herein, unless otherwise indicated. Top and bottom surfaces of style 47 insulator and top surface of insulator flange shall be parallel to each other and perpendicular to the center axis.

Type designation [1]	Dimensions (inches)						Y (degrees)	Insulators of following grades are available in appearances shown			
	D	F	G	J	K	M		L-3	L-4	L-5	L-6
NA--4601								C	N	CN	C
NA--4701	6 1/16	5 1 1/16	4 3/4	5 7/16	1 3/32	5 3/4	90	C	N	CN	C
NA--4702	6 1 3/16	6 5/16	5 1/2	6 5/16	1 1/16	5	60	C	N	CN	C

[1] Complete type designation will include additional symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (--).

[JAN–I–9]

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TYPE DESIGNATION TO BE IN THE FOLLOWING FORM

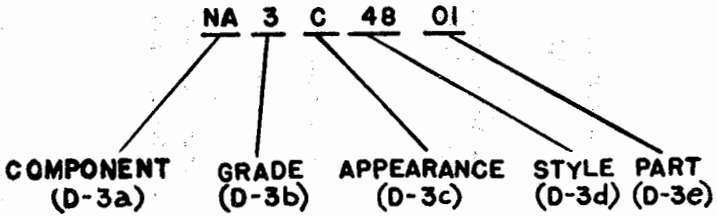


FIGURE 10—GLASS RADIO INSULATORS—LEAD-IN INSULATORS

Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

Type designation	Dimension (inches)	Insulators of following grades are available in appearances shown
	L	L-3
NA3C4801	6 9/16	C
NA3C4802	12 9/16	C
NA3C4803	15 9/16	C

[JAN–I–9]

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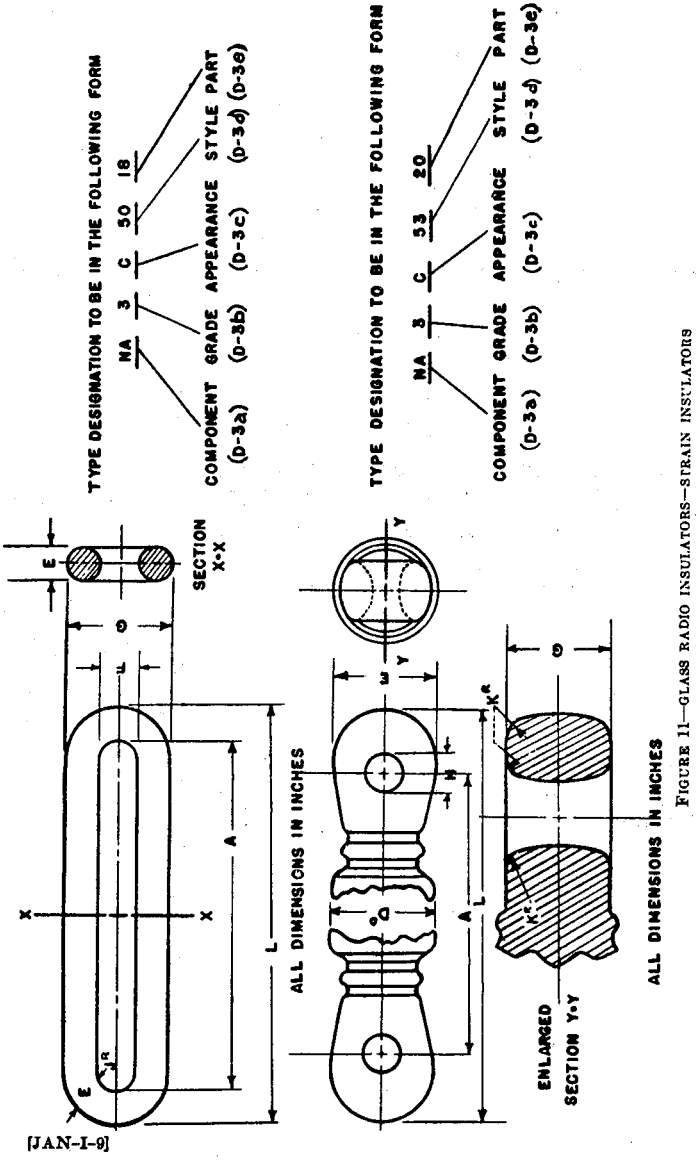


FIGURE 11—GLASS RADIO INSULATORS—STRAIN INSULATORS

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Figure 11—Glass Radio Insulators—Strain Insulators—(Continued)

Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

Type designation [1]	Dimensions (inches)									Breaking load in tension [2] (pounds)		Test wire Awg size [2]	Insulators of following grades are available in appearances shown	
	L	A	D	E	F	G	H (Approx.)	J	K (Approx.)	Minimum, any insulator	Average ultimate		L-3	L-4
NA--5018	2 7/8	2 1/4	-----	5/16	5/16	1 5/16	-----	5/32	-----	500	-----	7	C	N
NA--5026	3 7/8	3 1/4	-----	5/16	5/16	1 5/16	-----	5/32	-----	750	-----	2	C	N
NA--5050	7 1/4	6 1/4	-----	1/2	3/4	1 3/4	-----	3/8	-----	1,000	-----	2	C	N
NA--5320	3 5/8	2 1/2	1 1/8	1	-----	5/8	1/4	-----	5/32	100	300	7	C	N
NA--5342	7 1/2	5 1/4	1 7/8	1 11/16	-----	1 1/4	1/2	-----	1/4	375	750	2	C	N
NA--5380	12 1/4	10	1 7/8	1 7/8	-----	1 1/4	1/2	-----	1/4	375	750	2	C	N

[1] Complete type designation will include additional symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (--).

[2] Tensile strength tests shall be conducted with a 7-strand copper cable of the Awg size indicated looped through each end of the insulator.

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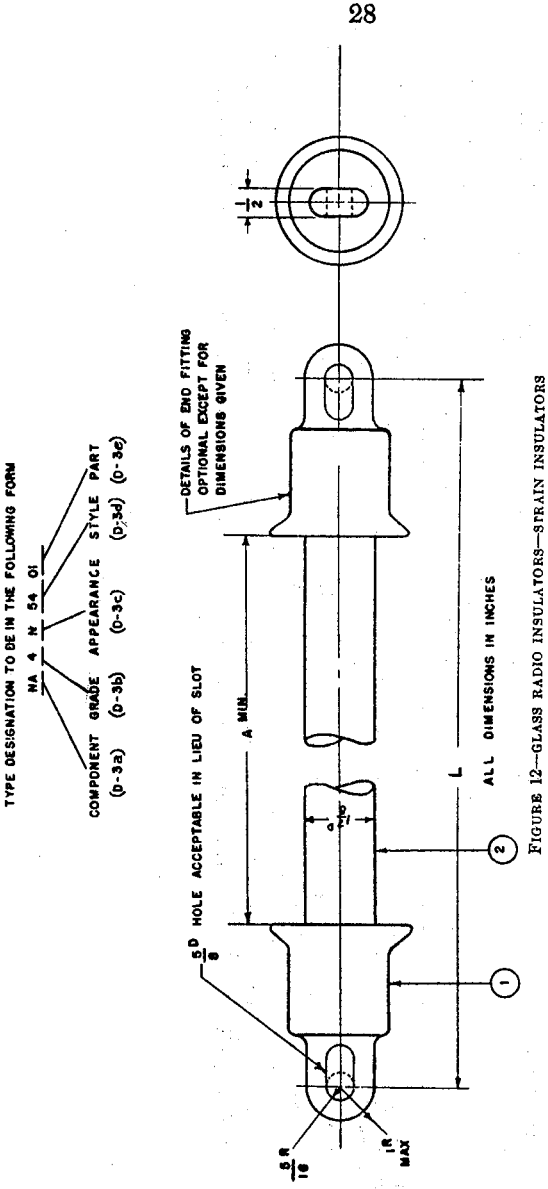


FIGURE 12—GLASS RADIO INSULATORS—STRAIN INSULATORS

[JAN–I–9]

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Figure 12—Glass Radio Insulator—Strain Insulators—(Continued)

BILL OF MATERIALS

Part No.	Name of part	Quantity required	Remarks
1	End fitting	2	Galvanized iron, Navy Dept Specification 46-I-8.
2	Insulator	1	Glass, Joint Army-Navy Specification JAN-1-10.

Tolerances of all dimensions of ceramic parts shall be as specified herein, unless otherwise indicated. Metal fittings to be finished before galvanizing.

Samples shall be tested to determine compliance with E-8.

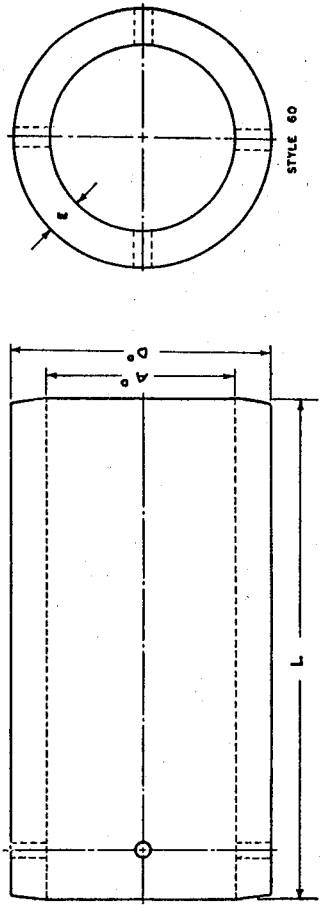
Method of securing metal parts to ceramic material optional.

Type designation[1]	Dimensions (inches)		Breaking load in tension [2] (pounds)		Insulators of following grades are available in appearances shown	
	L ±¼	A (Min)	Minimum, any insulator	Average ultimate	L-3	L-4
NA--5401	12	3⅞	3500	7000	C	N
NA--5402	18	9⅞	3500	7000	C	N
NA--5403	24	15⅞	3500	7000	C	N

[1] Complete type designation will include additional symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (--).

[2] Tensile strength tests shall be conducted with a ½-inch shackle through each hole.

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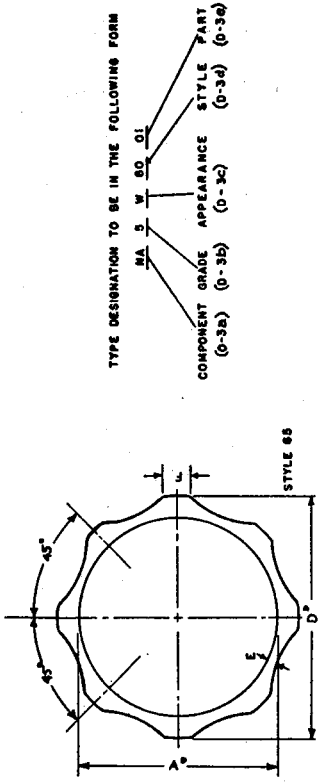


FIGURE 13—GLASS RADIO INSULATORS—TUBES, COIL AND MISCELLANEOUS (DIAMETER AND WALL THICKNESS)

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Figure 13—Glass Radio Insulators—Tubes, Coil and Miscellaneous (Diameter and Wall Thickness)—(Continued)

Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

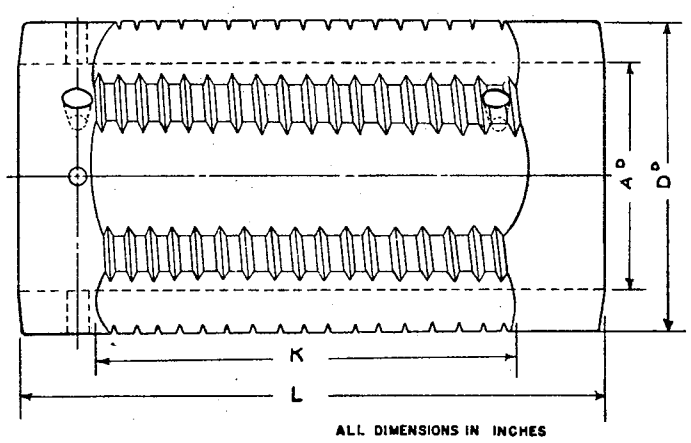
Type designation [1]	Dimensions (inches)				Insulators of following grades are available in appearances shown		
	A	D	E (Nominal)	F (Nominal)	L-3	L-5	L-6
NA--6001	5/16	1/2	3/32	—	C	W	CW
NA--6002	13/32	5/8	7/64	—	C	W	CW
NA--6003	1/2	3/4	1/8	—	C	W	CW
NA--6004	1 1/16	1	9/32	—	C	W	CW
NA--6005	3/4	1 1/8	3/16	—	C	W	CW
NA--6006	29/32	1 1/4	1 1/64	—	C	W	CW
NA--6007	1 1/8	1 1/2	3/16	—	C	W	CW
NA--6008	1 9/16	2	7/32	—	C	W	CW
NA--6009	2	2 1/2	1/4	—	C	W	CW
NA--6010	2 7/16	3	9/32	—	C	W	CW
NA--6011	3 9/16	4	1 1/32	—	C	W	CW
			Minimum / Maximum	Minimum / Maximum			
NA--6501	1 9/16	2	3/32 / 1/8	3/32	C	W	CW
NA--6502	2 1/16	2 1/2	1/8 / 3/16	1/4	C	W	CW
NA--6503	2 1/2	3	1/8 / 3/16	1/4	C	W	CW
NA--6504	3 1/4	4	5/32 / 7/32	3/8 / 1/2	C	W	CW
NA--6601	1 9/16	2	3/32 / 1/8	3/32	C	W	CW
NA--6602	2 1/16	2 1/2	1/8 / 3/16	1/4	C	W	CW
NA--6603	2 1/2	3	1/8 / 5/16	1/4	C	W	CW
NA--6604	3 1/4	4	5/32 / 7/32	3/8 / 1/2	C	W	CW

[1] Complete type designation will include additional symbols as shown in sketch. Finish and grade of material will be indicated by the symbols inserted for the hyphens (- -). Tubes have not been standardized with respect to number of coil-winding grooves per inch, location of lead holes, or length of coil form, but standard cross sections have been established. Under this specification, tubes are not required to carry individual part numbers but may be marked with the component and grade symbols of the type designation (Par. D-3).

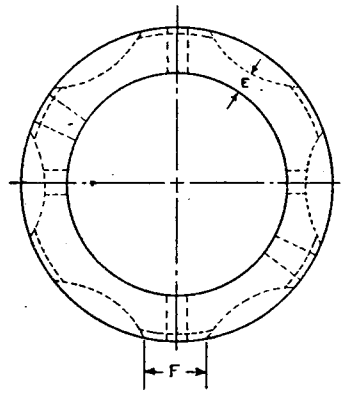
EXAMPLE.—NA3.

In specifying L, consideration should be given to end rings of Style 66. These rings facilitate manufacture of glass coil forms.

[JAN-I-9]



ALL DIMENSIONS IN INCHES



STYLE 66

(Face p. 30)

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Appendix

DESIGN CRITERIA

This section on design criteria was developed to be used as a reference for engineers and draftsmen in designing glass into their equipment.

The data compiled represents that which is considered good design practice and does not necessarily imply that closer tolerances on any important dimensions or special designs cannot be produced by special handling in some form or another.

It is definitely recommended that the following material be carefully considered by engineers and draftsmen in designing glass items, thus increasing production by reducing the possibility of waste, delay, and assembly difficulties. In order to insure the most desirable design and to facilitate procurement thereof at minimum cost and production time, it is recommended that designers consult the insulator manufacturer concerning specific design details applicable to the insulator under consideration.

1. For the purpose of this specification, glass insulators have been divided into 4 groups, each having distinct, characteristic manufacturing tolerances. Figures 1a, 2a, 3a, and 4a illustrate some of the types of ware obtainable in each characteristic, designated as characteristics A, B, C, and D.

2. *Holes—in characteristic A, B, or C ware.*—

2a. *General.*—Holes may be drilled, punched in the glass during the forming stage, or, in the case of pressed ware, can be pressed as a projection into the bottom and then ground open. This last type will be tapered from the inside to outside.

Tapped holes in these classes of ware are generally impracticable.

2b. *Tolerances.*—

2b (1). The tolerance on holes in these classes should be as follows:

Diameter (inch)	Tolerance (±)	
	Drilled or pressed	Punched
	Inch	*Inch*
From ⅛ to less than ¼	1/64	0.010
From ¼ to less than 1	1/32	----------

2b (2). For insulators fabricated with the use of jigs, the tolerance on hole-center spacing should be as follows:

Method	Tolerance on hole-center spacing (±)
	Inch
Drilled or pressed	1/32
Punched	0.020

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3. *Characteristic A ware (pressed ware).*—

3a. *General.*—Cup-shaped articles are preferred, although solid, noncavity ware can be made.

3b. *Side walls.*—The side-wall thickness should be not less than shown in Table Ia.

TABLE Ia

Diameter (inches)	Thickness (min.)
	Inch
Less than 2	3/32
From 2 to less than 4	4/32
4 or greater	5/16

Where ratio of depth to diameter is over 1, wall thickness should be increased. Nominal bottom thickness should be at least 50 percent greater than side-wall thickness.

3c. *Tolerances.*—

3c (1). Draft on inside should be not greater than the value shown in Table IIa.

TABLE IIa

Depth (inches)	Draft
	Degrees
Less than 3	2
From 3 to less than 5	3
From 5 to 7, inclusive	4

Draft on outside is desirable but not necessary, some undercuts being permissible on this surface. All designs should have radii at changes in direction of surface.

3c (2). *Diameter.*—The tolerance on diameter of this ware, including "out-of-round", should be not greater than the values shown in Table IIIa.

TABLE IIIa

Diameter (inches)	Tolerance (±)
	Inch
Less than 4	1/32
From 4 to less than 6	3/64
6 or greater	1/16

These tolerances also apply to solid, noncavity ware.

3c (3). *Side-wall thickness.*—Side walls that are approximately vertical should have a side-wall thickness tolerance of ±1/32 inch, exclusive of taper.

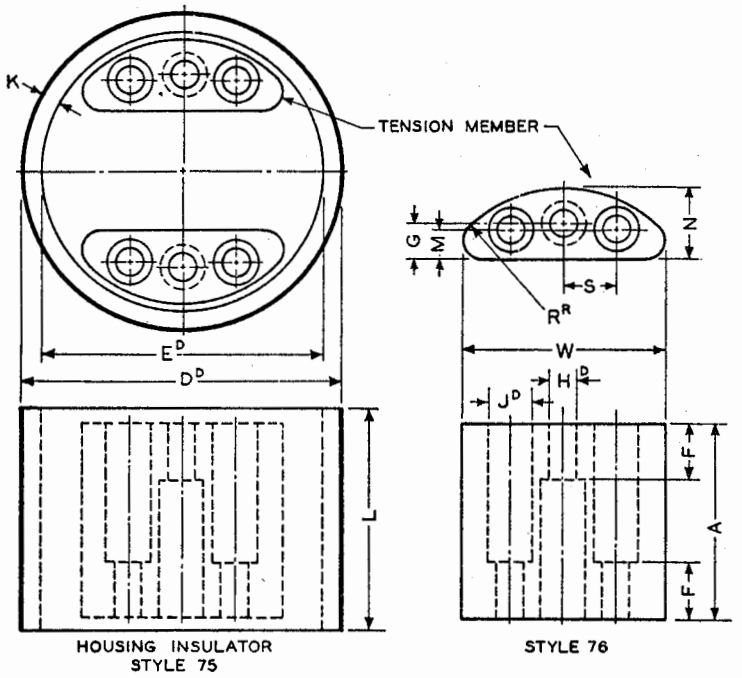
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TYPE DESIGNATION TO BE IN THE FOLLOWING FORM

NA 5 C 75 80

COMPONENT (D-3a) | GRADE (D-3b) | APPEARANCE (D-3c) | STYLE (D-3d) | PART (D-3e)



ALL DIMENSIONS IN INCHES

FIGURE 14—GLASS RADIO INSULATORS—SHELLS FOR TYPES CM80 TO CM90 FIXED MICA-DIELECTRIC CAPACITORS

(Figure 14—Continued)

Minimum and maximum diameters as shown include all allowances for out-of-round. Tolerances on all dimensions shall be as specified herein, unless otherwise indicated.

Type designation [1]	For mica capacitor	Dimensions (inches)				Insulators of following grades are available in appearances shown	
		L	D	E	K (min)	L-3	L-5
NS - - 7580	CM80	2¼ +0.050 −0.000	3⅛ ±1/16	2¾ ±1/16	3/16	C	CW
NS - - 7585	CM85	3 5/16 +0.065 −0.000	4½ ±⅛	4 ±⅛	¼	C	CW
NS - - 7590	CM90	5 1/16 +0.100 −0.000	4½ ±⅛	4 ±⅛	¼	C	CW

Type designation [1]	For mica capacitor	Dimensions (inches)									Insulators of following grades are available in appearance shown	
		A	F	G	H	J	M	N	R	S	W	L-3	L-5
NS - - 7630	CM80	2 +0.000 −0.040	9/16	3/8	17/64	7/16	5/16	2 3/32	1 1/32	1 7/32	2	C	CW
NS - - 7685	CM85	3 1/16 +0.000 −0.030	¾	3/8	17/64	7/16	5/16	1 31/16	1 1/16	¾	2⅞	C	CW
NS - - 7690	CM90	4 1/16 +0.000 −0.095	¾	3/8	17/64	7/16	5/16	1 31/16	1 1/16	¾	2⅞	C	CW

[1] Complete type designation will include additional symbols as shown in sketch. Appearance and grade of material will be indicated by the symbols inserted for the hyphens (-).

Ceramic part of Fixed Mica-Dielectric Capacitors Types CM80, CM85, and CM90, in accordance with the requirements of Joint Army-Navy Specification JAN-C-5, consists of 3 pieces: 1 housing insulator (Style 75) and 2 tension members (Style 76).

(Face p. 32)

3c (4). *Bottom thickness.*—The tolerance on closed-end thickness should be not greater than ±50 percent, unless otherwise specified. This will result in a corresponding change in the inside depth of the piece.

3c (5). *Over-all height or length.*—The tolerance on over-all height or length should be ±1/16 inch on mold-finished ware. This tolerance can be improved by finishing.

4. *Characteristic B ware (blown ware).*

4a. *General.*—For ware of this characteristic it is desirable that the section should be relatively close to circular, or, in the case of a rectangular section, the ratio of the $\frac{\text{length}}{\text{width}}$ of the cross section should be not greater than 2. The corners should have generous radii. The ratio of $\frac{\text{length}}{\text{diameter}}$ of cylinders should be not greater than 5. A considerably greater thickness should be allowed for the bottom of a closed cylinder than for the side walls. For shapes of varying diameter, the wall will be thicker than the average at points of constriction, and thinner than the average at points of expansion.

4b. *Tolerances.*—

4b (1). *Diameter—circular cross section.*—The tolerance on diameter of this ware, including "out-of-round", should be as shown in Table IVa.

4b (2). *Ellipticity.*—Ellipticity, or "out-of-round", should be determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient should not exceed 1.02.

4b (3). *Wall variation.*—For open-ended cylinders with a ratio of $\frac{\text{length}}{\text{diameter}}$ not greater than 5, the wall-thickness tolerance should be ± 33 percent; for a ratio greater than 5, the wall thickness tolerance should be ± 50 percent.

4b(4). With one end closed, the maximum thickness of the end should be not more than 50 percent greater than the maximum dimension for the side-wall thickness.

4b (5). For grooved cylinders, radio coil forms, and the like, a radius should be provided at the root of the grooves, and the intervening lands rounded. The diameter tolerance shown in Table IVa should be applied to the root diameter of the grooves.

TABLE IVa

Diameter (inches)	Tolerance (±)
	Inch
Less than 3	0.025
From 3 to less than 6	.032
From 6 to less than 10	.047
10 or greater	.063

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5. *Characteristic C ware (drawn tubing or rod).*

5a. *General.*—Tubing may be drawn either by machine or by hand. Hand drawing is employed for limited quantities of special sizes and also for special glass which is not readily adaptable to mechanical drawing equipment. Because of the difference in accuracy obtainable by the two methods, representative tolerances are listed for both methods.

5b. *Tolerances.*—

5b (1). *Outside diameter.*—Tolerances on outside diameter should be not greater than those shown on figure 1. These tolerances include allowances for ellipticity, or "out-of-round."

5b (1) a. *Solid rod.*—The tolerance on outside diameter of hand-drawn solid rod should be not greater than that shown for hand-drawn tubing on figure 1. For machine-drawn solid rod, the tolerance on outside diameter should be not greater than that given by the formula:

$$\text{Tolerance} = \pm (0.035\ D + 0.015)$$

where D is the outside diameter of the solid rod, and all dimensions are in inches.

NOTE.—Where the outside diameter of solid rod exceeds 1.25 inches, the tolerance value determined by the formula will not necessarily apply. Tolerances on solid rod of such diameters should be specified on the applicable figure or in the contract or order.

5b (2). *Wall thickness.*—The tolerance on wall thickness for machine-drawn tubing should be ±25 percent of nominal wall thickness; and for hand-drawn tubing, ±33 percent of nominal wall thickness. These values include allowances for nonconcentricity of bore and outside diameter. (See par. D–1e.)

5c. *Camber.*—The maximum permissible camber for machine-drawn tubing and rod should be 0.5 percent of the length of the tubing; and for hand-drawn tubing and rod, 0.75 percent of the length of the tubing.

6. *Characteristic D ware (cast or pressure-molded ware).*—

6a. *Tolerances.*—In general and unless otherwise specified, the notes below referring to tolerances apply to all dimensions of insulators. In the event that certain specific dimensions require tolerances closer than those required under "Critical Dimensions," they should be so indicated on the applicable figure.

6a (1). *Noncritical dimensions.*—All noncritical dimensions, which are indicated as such by common fractional dimensions on applicable figures or in contracts or orders, infer tolerances of ±2 percent of the dimensions involved, but not more than ±⅛ inch, on dimensions up to and including 12½ inches. On all dimensions above 12½ inches, a tolerance of ±1 percent should apply. In no case should any tolerance be less than ±1/64 inch.

6a (2) *Critical dimensions.*—All critical dimensions, which are indicated as such by decimal dimensions on applicable figures or in contracts or orders, infer tolerances of ±1 percent of the dimension involved, except as specified in paragraphs 6a (3) to 6a (11), but not less than ±0.010 inch, unless otherwise specified.

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6a (3). *Cylindrical shapes—outside and inside diameters.*—A tolerance of ±2 percent of the dimension involved, but not less than ±0.010 inch, should apply on all diameters up to and including 1 inch. Where the diameter is greater than 1 inch, ± (1 percent of the diameter involved +0.010 inch) should be used. All nominal wall thickness dimensions should bear a minimum tolerance of ±0.010 inch, unless otherwise specified.

Example:

Tube (inches)		*Tolerance (±) (inch)*	
O. D.	0.500	O. D.	0.010
I. D.	.125	I. D.	.010
O. D.	1.125	O. D.	.021
I. D.	.825	I. D.	.0165
O. D.	2.00	O. D.	.030
I. D.	1.750	I. D.	.0275

6a (4). *Lineal dimensions.*—All lineal dimensions representing overall lengths should have tolerances as shown in Table Va.

TABLE Va

Length (inches)	Tolerance (±)
	Inch
Less than 0.500	0.010
From 0.500 to less than 0.750	.012
From .750 to less than 1.000	.015
From 1.000 to less than 1.250	.020
From 1.250 to less than 1.500	.022
From 1.500 to less than 2.000	.025
From 2.000 to less than 2.500	.030
From 2.500 to less than 3.000	.035
From 3.000 to less than 4.000	.040
From 4.000 to less than 5.000	.045
From 5.000 to less than 6.000	.050
6.000 or greater	.055

6a (5). *Camber.*—The camber shown in table VIa should apply to all surfaces of all shapes. Figures 2, 3, and 4 should be used to establish the ratios listed in Table VIa.

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TABLE VIa

$\frac{d}{a}$	Maximum camber (percent of d) to be applied to surfaces
Less than 2	0.25
From 2 to less than 3	.30
From 3 to less than 5	.35
From 5 to less than 8	.40
From 8 to less than 13	.45
From 13 to less than 20	.50
From 20 to less than 30	.55
From 30 to less than 40	.60
From 40 to less than 50	.65

NOTE.—In determining "maximum camber" permitted, values of e (figs. 2, 3, and 4) less than 0.006 inch should be disregarded, unless otherwise specified.

For flat shapes, disks, plates, and the like, these limits should apply only where the thickness is equal to or greater than the thickness determined by the formulas given in paragraph 6a (10). Where the thickness is less than that determined as specified in paragraph 6a (10), the camber should be specified on the applicable figure or in the contract or order.

6a (6). *Wall thickness of tubes.*—The wall thickness of all cylindrical shapes should be not less than 8 percent of the outside diameter nor more than 20 percent or ⅜ inch, whichever is smaller, depending upon the length of the form as compared to the outside diameter. If the length of the form is greater than 3 times the diameter, but not over 6 times the diameter, it is desirable to increase the wall thickness as much as practicable up to 20 percent of the outside diameter. If the length is more than 6 times the outside diameter, the wall thickness should be approximately 10 percent of the outside diameter. Wall thicknesses of less than 1/32 inch or diameters smaller than 0.100 inch should be avoided.

NOTE.—See standard OD and ID dimensions, figure 13.

6a (7). *Ellipticity of tubes.*—The ellipticity is determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient should not exceed 1.02 and applies to cylindrical shapes having a wall thickness of at least 10 percent of the outside diameter. If the wall thickness at the thinnest section of the wall is less than 10 percent of the outside diameter, the same ratio should not exceed 1.03.

6a (8). *Hole centers.*—The standard tolerance for the dimension between centers of holes should be ±0.005 inch for all hole-center spacings less than 0.500 inch, and ±1 percent of the dimension involved for all hole-center spacings of 0.500 inch and greater. If pin gages are used for inspecting hole-center spacing, the design of the gage should be such as to meet the specific requirements. In using pin gages, considera-

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tion should be given to the tolerance between hole centers as well as hole-diameter tolerance.

6a (9). *Hole diameters.*—The standard tolerance for all hole diameters should be ± 0.005 inch for all diameters up to 0.500 inch, and ± 1 percent of the hole diameter for all diameters 0.500 inch and greater. Holes which are not perfect circles should have the same tolerance applied to the minor axis only.

6a (10). *Thickness.*—For flat shapes, disks, plates, and the like, the minimum thickness should be computed as follows:

Where the ratio of the maximum superficial dimension to the largest dimension at right angles to it is less than 5,

$$\text{Minimum thickness in inches} = \frac{\sqrt{A}}{10}$$

Where the same ratio is 5 or greater,

$$\text{Minimum thickness in inches} = \frac{\sqrt{A}}{8}$$

In both cases A is the outline area.

6a (11). *Parallel limits.*—Parallelism of surfaces is considered satisfactory if the thickness measured at any point is within the established thickness tolerance of the piece.

6b. *Extruded shapes.*—All holes perpendicular to the axis of extrusion are by the nature of the process somewhat elliptical. This fact should be taken into consideration by the designer.

6c. *Tapped holes.*—It is recommended that the use of tapped holes in glass parts be avoided if it is possible to provide other satisfactory methods of mounting. If tapped holes are specified, the following points should be taken into consideration:

6c (1). Due to the nature of material of Characteristic D, 6–32 or larger tapped holes should be specified. The largest practicable tapped holes should be used.

6c (2). All tapped holes should be slightly countersunk.

6c (3). It is recommended that the designer, wherever possible, increase the depth of the tapped hole below that required for the screw to an amount equivalent to at least 3 threads plus the original drill tip.

6c (4). Effective tapped holes should be a minimum of 6 threads and a maximum of 12 threads.

6d. *Wire groove threads.*—

6d (1). Grooved parts used for coil supports should be measured by a 3-wire method which includes 2 wires in adjacent grooves on one side and 1 wire in the groove between them on the other side of the coil form; and the specifications should include the wire diameter and the nominal diameter over wires with aforementioned diameter tolerances to apply.

6d (2). Where "V" threads are required, 90° is considered as standard. All others are considered as being special.

6d (3). A tolerance of $\pm 3°$ should be permitted on thread angle wherever possible.

6d (4). A slight radius is essential at the bottom of all grooves (instead of a sharp V).

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6e. *Miscellaneous design hints.*—

6e (1). Most of the points covered in paragraphs 2, 3, 4, and 5 also apply to ware of characteristic D covered in paragraph 6.

6e (2). *Walls.*—It is definitely desirable due to the fragility of the material to avoid all thin walls or thin barriers between recesses on these parts. The matter of thin sections is especially important at the end of a piece, that is the wall between a hole and the end of a tube or between a hole and a depression or an outside edge.

6e (3). *Thickness.*—It is recommended that all pressed parts be designed so as to maintain as nearly as possible a uniform thickness over the full area of the piece.

6e (4). *Holes.*—Exceedingly small holes and blind holes should be avoided where possible. Elongated holes, or preferably oversize round holes, should be specified where parts are to match corresponding holes in mating parts, so as to allow for variation in dimensions of the ceramic part.

6e (5). *Tapers and radii.*—Reasonable tapers and radii (drafts and fillets) should be specified for all depressions or raised bosses.

6e (6). *Edges.*—All edges where practicable should include a slight radius or bevel to prevent chipping during manufacturing operations and handling.

6e (7). *Flat mounting.*—Where parts are to be mounted on some flat surface, they should be so designed that they can be ground flat after firing to prevent breakage during the mounting operation.

6e (8). *Fastening of glass parts.*—Where glass parts are fastened rigidly to other parts of an assembly, it is recommended that a resilient material be provided to take up the differential expansion and thus prevent breakage of the glass. Suspension at 3 points or less is desirable.

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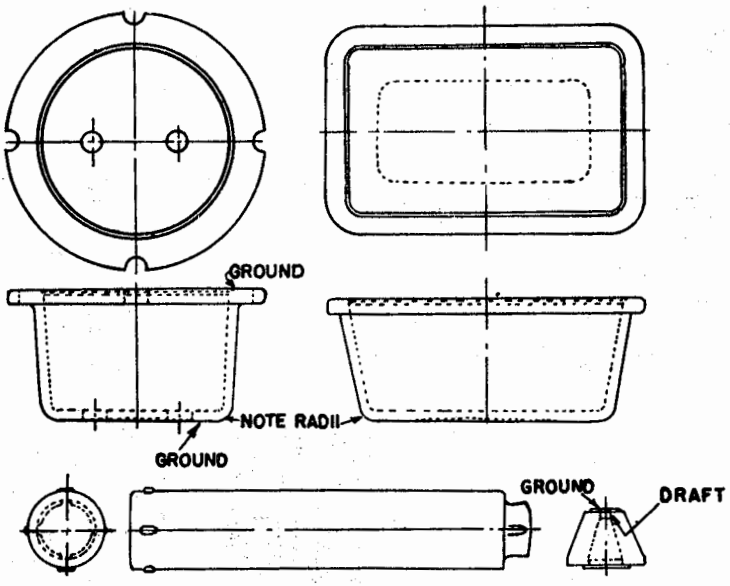


FIGURE 1A.—ARTICLES OF CHARACTERISTIC A.

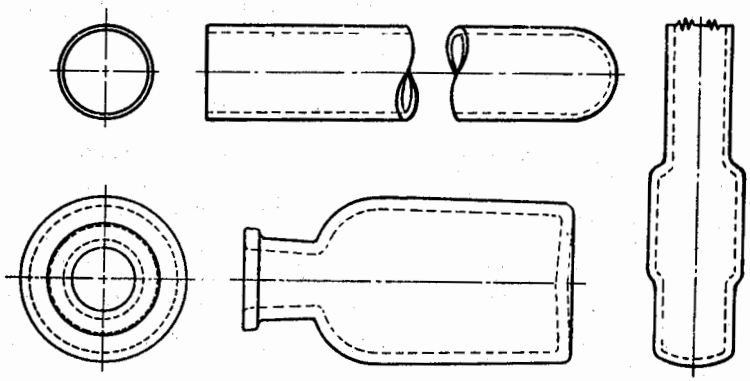


FIGURE 2A.—ARTICLES OF CHARACTERISTIC B.

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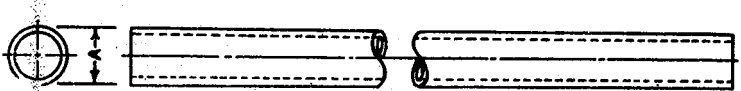


FIGURE 3A.—ARTICLES OF CHARACTERISTIC C.

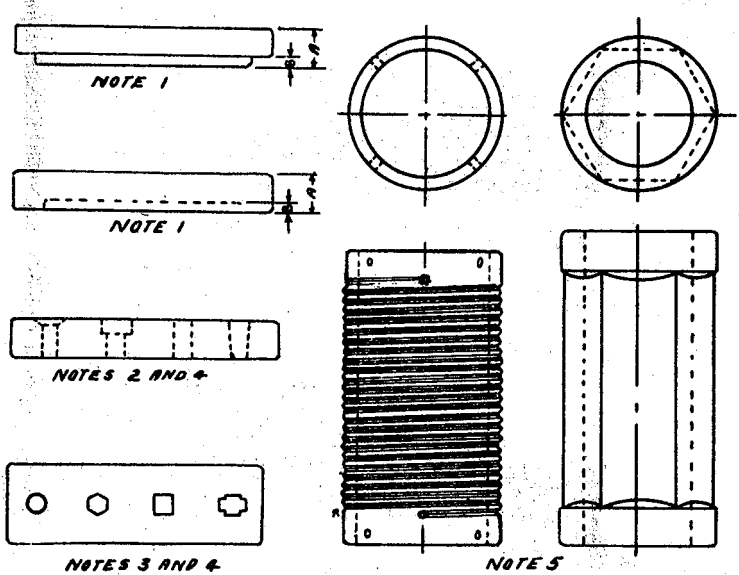


FIGURE 4A.—ARTICLES OF CHARACTERISTIC D.

NOTE 1.—Dimension B of projection or depression should be not greater than 25 percent of A.

NOTE 2.—Holes may be counterbored or countersunk. Threaded holes should be avoided, but, if used, should be not less than 6–32 and should be countersunk.

NOTE 3.—Holes should be perpendicular to major flat surface.

NOTE 4.—Depressions, projections, counterbores, etc., unless relatively shallow, should be on one surface only, leaving a flat opposite surface.

NOTE 5.—Ring (R) should be as wide as possible. Coil-form grooves should have a small radius.

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